STACKS, DISCRIMINANTS: LEMMA-UNIVERSAL-FINITE-SYNTOMIC-ETALE

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 2417-2423 of sources/stacks-discriminant/pinned-original-source.tex; proof span is lines 2425-2479, the complete author proof (53 body lines).

Editorial note. Navigation stubs: 11 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 3 of queue_discriminant.json (pb=2725, stmt 2417-2423, proof 2425-2479).

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Source reference. Original lemma 04GG: sources/stacks-discriminant/pinned-original-source.tex, source lines 2479–2479. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 03C3: sources/stacks-discriminant/pinned-original-source.tex, source lines 2479–2479. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0FKZ: sources/stacks-discriminant/pinned-original-source.tex, source lines 2326–2339. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0FK8: sources/stacks-discriminant/pinned-original-source.tex, source lines 2168–2185. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0FKY: sources/stacks-discriminant/pinned-original-source.tex, source lines 2282–2298. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0FL0: sources/stacks-discriminant/pinned-original-source.tex, source lines 2363–2368. Cited by the retained statement or proof but

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not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0FL1: sources/stacks-discriminant/pinned-original-source.tex, source lines 2384–2388. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0FK9: sources/stacks-discriminant/pinned-original-source.tex, source lines 2188–2195. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 06LN: sources/stacks-discriminant/pinned-original-source.tex, source lines 2479–2479. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 04HF: sources/stacks-discriminant/pinned-original-source.tex, source lines 2479–2479. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0476: sources/stacks-discriminant/pinned-original-source.tex, source lines 2479–2479. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *With notation as in Example 0FKZ consider the open subscheme $U'_d \subset X_d$ over which π_d is étale. Then U'_d is a dense subset of the open U_d of Lemma 0FL0.*

Proof. By exactly the same reasoning as in the proof of Lemma 0FL0, using Morphisms, Lemma 0476, there is a maximal open $U'_d \subset X_d$ over which π_d is étale. Moreover, since an étale morphism is syntomic, we see that $U'_d \subset U_d$. To finish the proof we have to show that $U'_d \subset U_d$ is dense. Let $u : \mathrm{Spec}(k) \to U_d$ be a morphism where k is a field. Let $B = k \otimes_{A_d} B_d$ as in the proof of Lemma 0FL1. We will show there is a local domain A' with residue field k and a finite syntomic A' algebra B' with $B = k \otimes_{A'} B'$ whose generic fibre is étale. Exactly as in the previous paragraph this will determine a morphism $\mathrm{Spec}(A') \to U_d$ which will map the generic point into U'_d and the closed point to u, thereby finishing the proof.

By Lemma 0FKY part (4) we can choose a presentation $B = k[x_1, \ldots, x_n]/(f_1, \ldots, f_n)$. Let d' be the maximum total degree of the polynomials $f_1, \ldots, f_n$. Let $Y_{n,d'} \to X_{n,d'}$ be as in Example 0FK8. By construction there is a morphism $u' : \mathrm{Spec}(k) \to X_{n,d'}$ such that

$$\mathrm{Spec}(B) \cong Y_{n,d'} \times_{X_{n,d'},u'} \mathrm{Spec}(k)$$

Denote $A = \mathcal{O}^h_{X_{n,d'},u'}$ the henselization of the local ring of $X_{n,d'}$ at the image of u'. Then we can write

$$Y_{n,d'} \times_{X_{n,d'}} \mathrm{Spec}(A) = Z \amalg W$$

with $Z \to \mathrm{Spec}(A)$ finite and $W \to \mathrm{Spec}(A)$ having empty closed fibre, see Algebra, Lemma 04GG part (13) or the discussion in More on Morphisms, Section 04HF. By Lemma 0FK9 the local ring A is regular (here we also use More on Algebra, Lemma 06LN) and the morphism $Z \to \mathrm{Spec}(A)$ is étale over the generic point of $\mathrm{Spec}(A)$ (because it is mapped to the generic point of $X_{d,n'}$). By construction $Z \times_{\mathrm{Spec}(A)} \mathrm{Spec}(k) \cong \mathrm{Spec}(B)$. This proves what we want except that the map

from residue field of A to k may not be an isomorphism. By Algebra, Lemma 03C3 there exists a flat local ring map $A \to A'$ such that the residue field of A' is k. If A' isn't a domain, then we choose a minimal prime $\mathfrak{p} \subset A'$ (which lies over the unique minimal prime of A by flatness) and we replace A' by $A'/\mathfrak{p}$. Set B' equal to the unique A'-algebra such that $Z \times_{\mathrm{Spec}(A)} \mathrm{Spec}(A') = \mathrm{Spec}(B')$. This finishes the proof. $\square$